

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-05

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TOPICS TO BE COVERED

1 Various Trigonometric identities

2 Related Problems

Trigonometric Functions of Sum and Difference of Two Angles

We know that,

$$\cos(x+y) = \cos x \cos y - \sin x \sin y \quad \text{--- (1)}$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y \quad \text{--- (2)}$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y \quad \text{--- (3)}$$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y \quad \text{--- (4)}$$

Adding (1) and (2), we get;

$$\begin{aligned} \cos(x+y) + \cos(x-y) &= \cancel{\cos x \cos y} - \cancel{\sin x \sin y} \\ &\quad + \cos x \cos y + \sin x \sin y \end{aligned}$$

$$\Rightarrow \boxed{\cos(x+y) + \cos(x-y) = 2\cos x \cos y}$$

Subtracting (1) and (2), we get

$$\begin{aligned} \cos(x+y) - \cos(x-y) &= \cancel{\cos x \cos y} - \sin x \sin y \\ &\quad - \cancel{\cos x \cos y} - \sin x \sin y \end{aligned}$$

$$\Rightarrow \boxed{\cos(x+y) - \cos(x-y) = -2\sin x \sin y}$$

Adding (3) and (4), we get;

$$\begin{aligned} \sin(x+y) + \sin(x-y) &= \sin x \cos y + \cancel{\cos x \sin y} \\ &\quad + \sin x \cos y - \cancel{\cos x \sin y} \end{aligned}$$

$$\Rightarrow \boxed{\sin(x+y) + \sin(x-y) = 2\sin x \cos y}$$

Subtracting ③ and ④, we get;

$$\sin(x+y) - \sin(x-y) = \cancel{\sin x \cos y} + \cos x \sin y - \cancel{\sin x \cos y} + \cos x \sin y$$

$$\Rightarrow \boxed{\sin(x+y) - \sin(x-y) = 2 \cos x \sin y.}$$

* Put $x=0$ in ②, we get,

$$\cos(0-y) = \cos 0 \cos y + \sin 0 \sin y.$$

$$\Rightarrow \boxed{\cos(-y) = \cos y} \quad (\cos(-\theta) = \cos \theta)$$

* Put $x=0$ in ④, we get,

$$\sin(0-y) = \sin 0 \cos y - \cos 0 \sin y.$$

$$\Rightarrow \boxed{\sin(-y) = -\sin y} \quad (\sin(-\theta) = -\sin \theta)$$

$$\text{Also } \tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = -\frac{\sin \theta}{\cos \theta}$$

$$\Rightarrow \boxed{\tan(-\theta) = -\tan \theta.}$$

QUESTION



If none of the angles x , y and $(x + y)$ is an odd multiple of $\frac{\pi}{2}$ then $\tan(x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

Proof: We know that

$$\tan(x + y) = \frac{\sin(x + y)}{\cos(x + y)}$$

$$\Rightarrow \tan(x + y) = \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y - \sin x \sin y}$$

Dividing the numerator & denominator of RHS by $\cos x \cos y$, we get;

$$\tan(x + y) = \frac{\left(\frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y} \right)}{\left(\frac{\cos x \cos y - \sin x \sin y}{\cos x \cos y} \right)}$$

$$\Rightarrow \tan(x + y) = \frac{\cancel{\sin x \cos y} + \cancel{\cos x \sin y}}{\cancel{\cos x \cos y} - \cancel{\sin x \sin y}} = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\Rightarrow \boxed{\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}}$$

$$* \tan(x+y) = \frac{\tan x + \tan(-y)}{1 - \tan x \tan(-y)}$$

$$\Rightarrow \boxed{\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}}$$

* Put $x=y$, we get;

$$\tan(x+x) = \frac{\tan x + \tan x}{1 - \tan x \times \tan x}$$

$$\Rightarrow \boxed{\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}}$$

QUESTION



If none of the angles x, y and $(x + y)$ is a multiple of π , then $\cot(x + y) = \frac{\cot x \cot y - 1}{\cot y + \cot x}$

$$\cot(x+y) = \frac{\cos(x+y)}{\sin(x+y)}$$

$$= \frac{\cos x \cos y - \sin x \sin y}{\sin x \cos y + \cos x \sin y} \quad (\text{Homework})$$

QUESTION



Prove that

$$\cos 2x = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

Proof: We know that

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

Put $x=y$, we get;

$$\cos(x+x) = \cos x \times \cos x - \sin x \times \sin x$$

$$\Rightarrow \cos 2x = \cos^2 x - \sin^2 x$$

$$\text{Also } \cos 2x = \cos^2 x - (1 - \cos^2 x)$$

$$\Rightarrow \cos 2x = 2\cos^2 x - 1$$

$$\text{Again } \cos 2x = 1 - \sin^2 x - \sin^2 x$$

$$\Rightarrow \cos 2x = 1 - 2\sin^2 x$$

$$\text{Further, } \cos 2x = \frac{\cos^2 x - \sin^2 x}{1}$$

$$\Rightarrow \cos 2x = \frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x}$$

$$\Rightarrow \cos 2x = \left(\frac{\cos^2 x - \sin^2 x}{\cos^2 x} \right) \left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right)$$

$$\Rightarrow \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

QUESTION



Prove that

$$\sin 2x = 2 \sin x \cos x = \frac{2 \tan x}{1 + \tan^2 x} \quad x \neq n\pi + \frac{\pi}{2} \text{ where } n \text{ is an integer}$$

Proof: We know that

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

Put $x = y$, we get,

$$\sin(x+x) = \sin x \cos x + \cos x \sin x$$

$$\Rightarrow \sin 2x = 2 \sin x \cos x$$

$$\begin{aligned} \text{Also, } \sin 2x &= \frac{2 \sin x \cos x}{1} = \frac{2 \sin x \cos x}{\cos^2 x + \sin^2 x} \\ \Rightarrow \sin 2x &= \left(\frac{2 \sin x \cancel{\cos x}}{\cos^2 x} \right) \left(\frac{\cos^2 x + \sin^2 x}{\cos^2 x} \right) = \frac{2 \tan x}{1 + \tan^2 x} \end{aligned}$$

$$\Rightarrow \sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$$

QUESTION

Prove the following:

(a) $\sin 3x = 3\sin x - 4\sin^3 x$

(b) $\cos 3x = 4\cos^3 x - 3\cos x$

(c) $\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$ if $3x \neq n\pi + \frac{\pi}{2}$, where $n \in \mathbb{Z}$

Homework

^(a)
Proof: Here $\sin 3x = \sin(2x+x)$

$$\Rightarrow \sin 3x = \sin 2x \cos x + \cos 2x \sin x$$

$$\Rightarrow \sin 3x = 2 \sin x \cos x \times \cos x + (1 - 2 \sin^2 x) \sin x$$

$$\Rightarrow \sin 3x = 2 \sin x \cos^2 x + \sin x - 2 \sin^3 x$$

$$\Rightarrow \sin 3x = 2 \sin x (1 - \sin^2 x) + \sin x - 2 \sin^3 x$$

$$\Rightarrow \sin 3x = 2 \sin x - 2 \sin^3 x + \sin x - 2 \sin^3 x$$

$$\Rightarrow \boxed{\sin 3x = 3 \sin x - 4 \sin^3 x}$$

(b) Here $\cos 3x = \cos(2x+x)$

$$\Rightarrow \cos 3x = \cos 2x \cos x - \sin 2x \sin x$$

$$\Rightarrow \cos 3x = (2 \cos^2 x - 1) \cos x - 2 \sin^2 x \cos x$$

$$\Rightarrow \cos 3x = 2 \cos^3 x - \cos x - 2(1 - \cos^2 x) \cos x$$

$$\Rightarrow \cos 3x = 2 \cos^3 x - \cos x - 2 \cos x + 2 \cos^3 x$$

$$\Rightarrow \boxed{\cos 3x = 4 \cos^3 x - 3 \cos x}$$

QUESTION

Prove the following:

$$(i) \quad \cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2} \quad \checkmark$$

$$(ii) \quad \cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2} \quad \checkmark$$

$$(iii) \quad \sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2} \quad \checkmark$$

$$(iv) \quad \sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2} \quad \checkmark$$

yaar kar lena.

QUESTION

The value of $\sin(45^\circ + \theta) - \cos(45^\circ - \theta)$ is

- ☒ 1 $2 \cos \theta$
- ☒ 2 $2 \sin \theta$
- ☒ 3 1
- ☒ 4 0

$$\begin{aligned}
 &= (\sin 45^\circ \cos \theta + \cos 45^\circ \sin \theta) - (\cos 45^\circ \cos \theta + \sin 45^\circ \sin \theta) \\
 &= \frac{\cos \theta}{\sqrt{2}} + \frac{\sin \theta}{\sqrt{2}} - \frac{\cos \theta}{\sqrt{2}} - \frac{\sin \theta}{\sqrt{2}} \\
 &= 0.
 \end{aligned}$$

QUESTION

If $\sin \theta + \cos \theta = 1$, then the value of $\sin 2\theta$ is equal to

- ☒ 1
- ☒ $\frac{1}{2}$
- ☒ 0
- ☒ -1

$$\begin{aligned}
 \text{(Sol)} \quad & \sin \theta + \cos \theta = 1 \\
 \Rightarrow & (\sin \theta + \cos \theta)^2 = 1^2 \\
 \Rightarrow & \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 1 \\
 \Rightarrow & \cancel{1} + \sin 2\theta = \cancel{1} \\
 \Rightarrow & \boxed{\sin 2\theta = 0}
 \end{aligned}$$

QUESTION

The value of $\sin(40^\circ + \theta) \cos(10^\circ + \theta) - \cos(40^\circ + \theta) \sin(10^\circ + \theta)$

$$= \sin[(40^\circ + \theta) - (10^\circ + \theta)]$$

$$= \sin(30^\circ) = \frac{1}{2}$$

☒ 1☒ 2 $\frac{1}{2}$ ☐ 3

0

☐ 4 $\frac{1}{4}$

QUESTION

The value of $\cos 35^\circ + \cos 85^\circ + \cos 155^\circ$ is

0

$\frac{1}{\sqrt{3}}$

$\frac{1}{\sqrt{2}}$

$\cos 275^\circ$

$$(Sol) \cos 35^\circ + \cos 85^\circ + \cos 155^\circ$$

$$= [\cos(60^\circ - 25^\circ) + \cos(60^\circ + 25^\circ)] + \cos 155^\circ.$$

$$= 2 \cos 60^\circ \cos 25^\circ + \cos 155^\circ.$$

$$= 2 \times \frac{1}{2} \cos 25^\circ + \cos 155^\circ$$

$$= \cos 25^\circ + \cos 155^\circ$$

$$= \cos(90^\circ - 65^\circ) + \cos(90^\circ + 65^\circ)$$

$$= 2 \cos 90^\circ \cos 65^\circ = 2 \times 0 \times \cos 65^\circ = 0.$$

QUESTION

The value of $\frac{1 - \tan^2 15^\circ}{1 + \tan^2 15^\circ}$ is

~~1~~

1

~~2~~

$\sqrt{3}$

~~3~~

$\frac{\sqrt{3}}{2}$

~~4~~

2

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$\frac{1 - \tan^2(15^\circ)}{1 + \tan^2(15^\circ)} = \cos(2 \times 15^\circ) = \cos(30^\circ) = \frac{\sqrt{3}}{2}$$

QUESTION

If $\alpha + \beta = \frac{\pi}{4}$, then the value of $(1 + \tan \alpha)(1 + \tan \beta)$ is

- 1** 1 *(Homework Problem)*
- 2** 2
- 3** -2
- 4** Not defined

QUESTION

If $\tan A = \frac{1}{2}$, $\tan B = \frac{1}{3}$, then $\tan (2A + B)$ is equal to

1

1

*(Homework Problem)***2**

2

3

3

4

4

QUESTION

The range of the function $f(x) = \frac{1}{4\cos x - 3}$ is

1 $\left(-\frac{1}{7}, 1\right)$

Homework Problem.

2 $\left[-\frac{1}{7}, 1\right]$

3 $\left(-\infty, -\frac{1}{7}\right] \cup [1, \infty)$

4 None of these

QUESTION(JEE-MAINS 2023)

The value of $36(4 \cos^2 9^\circ - 1)(4 \cos^2 27^\circ - 1)(4 \cos^2 81^\circ - 1)(4 \cos^2 243^\circ - 1)$ is:

- ☒ 1 54
- ☒ 2 18
- ☒ 3 27
- ☒ 4 36

(Sol) $4 \cos^2 \theta - 1 = 4(1 - \sin^2 \theta) - 1$
 $\Rightarrow 4 \cos^2 \theta - 1 = 4 - 4 \sin^2 \theta - 1$
 $\Rightarrow 4 \cos^2 \theta - 1 = \frac{(3 - 4 \sin^2 \theta) \sin \theta}{\sin \theta}$
 $\Rightarrow 4 \cos^2 \theta - 1 = \frac{3 \sin \theta - 4 \sin^3 \theta}{\sin \theta}$
 $\Rightarrow 4 \cos^2 \theta - 1 = \frac{\sin 3\theta}{\sin \theta}$

$= 36 \times \frac{\cancel{\sin 27^\circ}}{\cancel{\sin 9^\circ}} \times \frac{\cancel{\sin 81^\circ}}{\cancel{\sin 27^\circ}} \times \frac{\cancel{\sin 243^\circ}}{\cancel{\sin 81^\circ}} \times \frac{\cancel{\sin 729^\circ}}{\cancel{\sin 243^\circ}}$
 $= 36 \frac{\sin 729^\circ}{\sin 9^\circ} = 36 \times \frac{\sin(2 \times 360^\circ + 9^\circ)}{\sin 9^\circ}$
 $= 36 \frac{\cancel{\sin 9^\circ}}{\cancel{\sin 9^\circ}} = 36$

QUESTION(JEE-MAINS 2026)

The value of $\operatorname{cosec} 10^\circ - \sqrt{3} \sec 10^\circ$ is:

~~1~~

1

~~2~~

2

~~3~~

4

~~4~~

None of these

$$= \left(\frac{1}{\sin 10^\circ} - \frac{\sqrt{3}}{\cos 10^\circ} \right)$$

$$= \left(\frac{\cos 10^\circ - \sqrt{3} \sin 10^\circ}{\sin 10^\circ \cos 10^\circ} \right)$$

$$= \frac{2 \left(\frac{1}{2} \cos 10^\circ - \frac{\sqrt{3}}{2} \sin 10^\circ \right)}{\sin 10^\circ \cos 10^\circ}$$

$$= 2 \left(\frac{\sin 30^\circ \cos 10^\circ - \cos 30^\circ \sin 10^\circ}{\sin 10^\circ \cos 10^\circ} \right)$$

$$= \frac{2 \sin(30^\circ - 10^\circ)}{\sin 10^\circ \cos 10^\circ}$$

$$= \frac{2 \sin 20^\circ \times 2}{2 \sin 10^\circ \cos 10^\circ}$$

$$= \frac{4 \sin 10^\circ}{\sin 20^\circ}$$

Thank
You